SAMPLE: SYNTHETIC DATA Mini-grid site performance GI-MSP-2026-09-14-2C52143B

GRIDINTEL · MINI-GRID INTELLIGENCE

Mini-grid site performance

The sites side by side, then each site: energy flow, PV, battery and generator, supply and interruptions, sales and collection.

SAMPLE: SYNTHETIC DATA

This report is a sample. It shows the format and the methods GridIntel would deliver; it is not a report of real results. Every figure in it is calculated from a designed demonstration dataset, and none describes a real network, site, customer or organization.

A designed demonstration dataset for September 2026. It describes no real network, customer, meter reading, outage, bill or payment.

FOR
Mini-grid operator

RECORDS OF
Demonstration Mini-grid Operator (synthetic)

PERIOD
14 September 2026

REPORT ID
GI-MSP-2026-09-14-2C52143B

FIGURES COMPUTED
20 Oct 2026, 10:10 WAT, from data as of 15 Sep 2026, 00:00 WAT

DEFINITION
Version 1

DATA FINGERPRINT (SHA-256)
b0167ca08ac57457af3bea882205afb118f40ea88543b10d1bbe5fea52b64234

IN BRIEF

SITES IN THE REGISTER
4 • C
OK · Calculated

AVAILABILITY, DRY RIVER
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Every recorded interruption counted, planned or not.

AVAILABILITY, GRANITE HILL
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Every recorded interruption counted, planned or not.

AVAILABILITY, PALM LANDING
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Every recorded interruption counted, planned or not.

AVAILABILITY, TAMARIND CROSS
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Every recorded interruption counted, planned or not.

RENEWABLE FRACTION, DRY RIVER
46.2% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

RENEWABLE FRACTION, GRANITE HILL
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

RENEWABLE FRACTION, PALM LANDING
50.9% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

RENEWABLE FRACTION, TAMARIND CROSS
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

Sales and collection are not given for a day: Collection efficiency and revenue per connection compare money billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is used, and payments arrive on days of their own, so for a single day they are not on one basis. They are given in the month's report.

CONTENTS

1. The sites side by side
2. Dry River
3. Granite Hill
4. Palm Landing
5. Tamarind Cross
6. Method, sources and how to reproduce this report
7. Regulatory references
8. Glossary and key

1 The sites side by side

How each site's plant, supply and sales compare.

Plant and energy, by site

	CONNECTIONS	RENEWABLE FRACTION	PERFORMANCE RATIO	PV CURTAILED	FUEL PER KWH	AVAILABILITY
Dry River 45 kWp PV · 90 kWh battery · 60 kVA diesel generator	300 • C	46.2% • C	76.0% • C	0.0% ≈ C	0.51 L/kWh • C	100.0% • C
Granite Hill 60 kWp PV · 150 kWh battery · no generator	130 • C	100.0% • C	23.7% • C	70.0% ≈ C	No generator	100.0% • C
Palm Landing 80 kWp PV · 200 kWh battery · 40 kVA diesel generator	393 • C	50.9% • C	77.0% • C	0.0% ≈ C	0.35 L/kWh • C	100.0% • C
Tamarind Cross 100 kWp PV · 240 kWh battery · 50 kVA diesel generator	465 • C	100.0% • C	71.0% • C	10.1% ≈ C	*not computable* ∅	100.0% • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

Sales and collection: not available for a day. Not available for a day. Collection efficiency and revenue per connection compare money billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is used, and payments arrive on days of their own, so for a single day they are not on one basis. They are given in the month's report.

Most fuel per kWh: Dry River. Dry River's generator burns the most fuel for each kWh it makes, and ran the longest. Rule: Of the sites with a generator, the one with the highest fuel per kWh, when it is at least 20% above the lowest.

PV held back for want of demand: Granite Hill. Granite Hill took less than half of the PV energy available to it. Its array is not underperforming: it is larger than its demand. Rule: At least 50% of the PV energy available to the site was curtailed.

2 Dry River

Dry River's energy, plant, supply; commissioned 2019-11-01.

Energy

RENEWABLE FRACTION, DRY RIVER
46.2% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

SITE LOSS, DRY RIVER
insufficient data !
Insufficient data · Derived · gridintel.site.reference 0.1.0
Energy generated that reached no customer meter, as a share of energy generated.

DISTRIBUTION LOSS, DRY RIVER
insufficient data !
Insufficient data · Derived · gridintel.site.reference 0.1.0
Share of energy sent out.

Energy flow, Dry River

From generation to customer meters, top to bottom. Each line keeps its own origin: a residual is shown as one.

	IN THE CHAIN	ENERGY
PV generation	Adds	136 kWh • C
Generator generation	Adds	158 kWh • C
Energy generated	Equals	295 kWh • C
Net into storage Energy into the battery less energy out of it: storage loss and any change in what is stored.	Less	-16 kWh • C
Station use A residual: the powerhouse's own use and anything unaccounted for.	Less	7 kWh • D
Energy sent out Metered where the powerhouse hands over to the network: the energy available to customers.	Equals	303 kWh • C
Distribution loss A residual: technical loss and unmetered use together.	Less	*insufficient data* !
Energy delivered What customer meters recorded.	Equals	*insufficient data* !

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

294 of 300 connections have a meter that recorded the whole period; 0 are unmetered and 6 incomplete.

Solar PV

PERFORMANCE RATIO, DRY RIVER
76.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Counts curtailment as a loss, as IEC 61724 does.

PERFORMANCE RATIO BEFORE CURTAILMENT, DRY RIVER
76.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Adds the curtailment estimate back: what the array could do.

PV CURTAILED, DRY RIVER
0.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Of the PV energy that was available.

IN-PLANE IRRADIATION, DRY RIVER
4.0 kWh/m² • C
OK · Calculated · gridintel.site.reference 0.1.0
Measured on site; the input to the performance ratio.

SPECIFIC YIELD, DRY RIVER
3.0 kWh/kWp • C
OK · Calculated · gridintel.site.reference 0.1.0

CURTAILMENT, DRY RIVER
0 kWh ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0
The controller's estimate. Not subtracted from metered generation, which is already net of it.

Battery (90 kWh, 40 kW, lithium iron phosphate)

EQUIVALENT FULL CYCLES, DRY RIVER
0.7 • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy out of the battery over its nameplate capacity.

ROUND-TRIP EFFICIENCY, DRY RIVER
69.7% ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0

STORAGE LOSS, DRY RIVER
15 kWh ≈ D
Estimated inputs · Derived · gridintel.site.reference 0.1.0
The change in stored energy is estimated from state of charge and nameplate capacity.

STATE OF CHARGE, MEAN, DRY RIVER
47.3% • C
OK · Calculated · gridintel.site.reference 0.1.0

STATE OF CHARGE, LOWEST, DRY RIVER
20.9% • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY INTO STORAGE, DRY RIVER
51 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY OUT OF STORAGE, DRY RIVER
67 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

Generator (60 kVA, 48 kW, diesel)

FUEL PER KWH, DRY RIVER
0.51 L/kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Litres of fuel for each kWh the generator made.

RUNNING HOURS, DRY RIVER
11.0 h • C
OK · Calculated · gridintel.site.reference 0.1.0
The advance of the hour meter over the period.

FUEL USED, DRY RIVER
80 L • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR GENERATION, DRY RIVER
158 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR'S SHARE OF ENERGY, DRY RIVER
53.8% • C
OK · Calculated · gridintel.site.reference 0.1.0

Supply

AVAILABILITY, DRY RIVER
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

HOURS OF SUPPLY PER DAY, DRY RIVER
24.0 h/day • C
OK · Calculated · gridintel.site.reference 0.1.0

INTERRUPTIONS PER CONNECTION, DRY RIVER
0.0 • C
OK · Calculated · gridintel.site.reference 0.1.0

AVAILABILITY WITHOUT PLANNED INTERRUPTIONS, DRY RIVER
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

Interruptions, Dry River

Every interruption recorded at the site in the period, in order.

> No interruption is recorded at Dry River in the period.

Sales and collection: not available for a day. Not available for a day. Collection efficiency and revenue per connection compare money billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is used, and payments arrive on days of their own, so for a single day they are not on one basis. They are given in the month's report.

3 Granite Hill

Granite Hill's energy, plant, supply; commissioned 2024-09-01.

Energy

RENEWABLE FRACTION, GRANITE HILL
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

SITE LOSS, GRANITE HILL
16.4% • D
OK · Derived · gridintel.site.reference 0.1.0
Energy generated that reached no customer meter, as a share of energy generated.

DISTRIBUTION LOSS, GRANITE HILL
5.0% • D
OK · Derived · gridintel.site.reference 0.1.0
Share of energy sent out.

Energy flow, Granite Hill

From generation to customer meters, top to bottom. Each line keeps its own origin: a residual is shown as one.

	IN THE CHAIN	ENERGY
PV generation	Adds	63 kWh • C
Generator generation	Adds	0 kWh • C
Energy generated	Equals	63 kWh • C
Net into storage Energy into the battery less energy out of it: storage loss and any change in what is stored.	Less	2 kWh • C
Station use A residual: the powerhouse's own use and anything unaccounted for.	Less	6 kWh • D
Energy sent out Metered where the powerhouse hands over to the network: the energy available to customers.	Equals	55 kWh • C
Distribution loss A residual: technical loss and unmetered use together.	Less	3 kWh • D
Energy delivered What customer meters recorded.	Equals	53 kWh • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

130 of 130 connections have a meter that recorded the whole period; 0 are unmetered and 0 incomplete.

Solar PV

PERFORMANCE RATIO, GRANITE HILL
23.7% • C
OK · Calculated · gridintel.site.reference 0.1.0
Counts curtailment as a loss, as IEC 61724 does.

PERFORMANCE RATIO BEFORE CURTAILMENT, GRANITE HILL
79.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Adds the curtailment estimate back: what the array could do.

PV CURTAILED, GRANITE HILL
70.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Of the PV energy that was available.

IN-PLANE IRRADIATION, GRANITE HILL
4.4 kWh/m² • C
OK · Calculated · gridintel.site.reference 0.1.0
Measured on site; the input to the performance ratio.

SPECIFIC YIELD, GRANITE HILL
1.0 kWh/kWp • C
OK · Calculated · gridintel.site.reference 0.1.0

CURTAILMENT, GRANITE HILL
146 kWh ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0
The controller's estimate. Not subtracted from metered generation, which is already net of it.

Battery (150 kWh, 40 kW, lithium iron phosphate)

EQUIVALENT FULL CYCLES, GRANITE HILL
0.2 • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy out of the battery over its nameplate capacity.

ROUND-TRIP EFFICIENCY, GRANITE HILL
90.0% ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0

STORAGE LOSS, GRANITE HILL
3 kWh ≈ D
Estimated inputs · Derived · gridintel.site.reference 0.1.0
The change in stored energy is estimated from state of charge and nameplate capacity.

STATE OF CHARGE, MEAN, GRANITE HILL
93.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

STATE OF CHARGE, LOWEST, GRANITE HILL
83.2% • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY INTO STORAGE, GRANITE HILL
26 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY OUT OF STORAGE, GRANITE HILL
24 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

Generator. Granite Hill has no generator in the register.

Supply

AVAILABILITY, GRANITE HILL
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

HOURS OF SUPPLY PER DAY, GRANITE HILL
24.0 h/day • C
OK · Calculated · gridintel.site.reference 0.1.0

INTERRUPTIONS PER CONNECTION, GRANITE HILL
0.0 • C
OK · Calculated · gridintel.site.reference 0.1.0

AVAILABILITY WITHOUT PLANNED INTERRUPTIONS, GRANITE HILL
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

Interruptions, Granite Hill

Every interruption recorded at the site in the period, in order.

No interruption is recorded at Granite Hill in the period.

Sales and collection: not available for a day. Not available for a day. Collection efficiency and revenue per connection compare money billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is used, and payments arrive on days of their own, so for a single day they are not on one basis. They are given in the month's report.

4 Palm Landing

Palm Landing's energy, plant, supply; commissioned 2021-06-01.

Energy

RENEWABLE FRACTION, PALM LANDING
50.9% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

SITE LOSS, PALM LANDING
5.0% • D
OK · Derived · gridintel.site.reference 0.1.0
Energy generated that reached no customer meter, as a share of energy generated.

DISTRIBUTION LOSS, PALM LANDING
14.0% • D
OK · Derived · gridintel.site.reference 0.1.0
Share of energy sent out.

Energy flow, Palm Landing

From generation to customer meters, top to bottom. Each line keeps its own origin: a residual is shown as one.

	IN THE CHAIN	ENERGY
PV generation	Adds	149 kWh • C
Generator generation	Adds	144 kWh • C
Energy generated	Equals	293 kWh • C
Net into storage Energy into the battery less energy out of it: storage loss and any change in what is stored.	Less	-38 kWh • C
Station use A residual: the powerhouse's own use and anything unaccounted for.	Less	7 kWh • D
Energy sent out Metered where the powerhouse hands over to the network: the energy available to customers.	Equals	324 kWh • C
Distribution loss A residual: technical loss and unmetered use together.	Less	45 kWh • D
Energy delivered What customer meters recorded.	Equals	279 kWh • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

417 of 417 connections have a meter that recorded the whole period; 0 are unmetered and 0 incomplete.

Solar PV

PERFORMANCE RATIO, PALM LANDING
77.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Counts curtailment as a loss, as IEC 61724 does.

PERFORMANCE RATIO BEFORE CURTAILMENT, PALM LANDING
77.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Adds the curtailment estimate back: what the array could do.

PV CURTAILED, PALM LANDING
0.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Of the PV energy that was available.

IN-PLANE IRRADIATION, PALM LANDING
2.4 kWh/m² • C
OK · Calculated · gridintel.site.reference 0.1.0
Measured on site; the input to the performance ratio.

SPECIFIC YIELD, PALM LANDING
1.9 kWh/kWp • C
OK · Calculated · gridintel.site.reference 0.1.0

CURTAILMENT, PALM LANDING
0 kWh ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0
The controller's estimate. Not subtracted from metered generation, which is already net of it.

Battery (200 kWh, 60 kW, lithium iron phosphate)

EQUIVALENT FULL CYCLES, PALM LANDING
0.6 • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy out of the battery over its nameplate capacity.

ROUND-TRIP EFFICIENCY, PALM LANDING
65.3% ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0

STORAGE LOSS, PALM LANDING
25 kWh ≈ D
Estimated inputs · Derived · gridintel.site.reference 0.1.0
The change in stored energy is estimated from state of charge and nameplate capacity.

STATE OF CHARGE, MEAN, PALM LANDING
34.2% • C
OK · Calculated · gridintel.site.reference 0.1.0

STATE OF CHARGE, LOWEST, PALM LANDING
18.6% • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY INTO STORAGE, PALM LANDING
73 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY OUT OF STORAGE, PALM LANDING
111 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

Generator (40 kVA, 32 kW, diesel)

FUEL PER KWH, PALM LANDING
0.35 L/kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Litres of fuel for each kWh the generator made.

RUNNING HOURS, PALM LANDING
6.0 h • C
OK · Calculated · gridintel.site.reference 0.1.0
The advance of the hour meter over the period.

FUEL USED, PALM LANDING
50 L • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR GENERATION, PALM LANDING
144 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR'S SHARE OF ENERGY, PALM LANDING
49.1% • C
OK · Calculated · gridintel.site.reference 0.1.0

Supply

AVAILABILITY, PALM LANDING
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

HOURS OF SUPPLY PER DAY, PALM LANDING
24.0 h/day • C
OK · Calculated · gridintel.site.reference 0.1.0

INTERRUPTIONS PER CONNECTION, PALM LANDING
0.0 • C
OK · Calculated · gridintel.site.reference 0.1.0

AVAILABILITY WITHOUT PLANNED INTERRUPTIONS, PALM LANDING
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

Interruptions, Palm Landing

Every interruption recorded at the site in the period, in order.

> No interruption is recorded at Palm Landing in the period.

Sales and collection: not available for a day. Not available for a day. Collection efficiency and revenue per connection compare money billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is used, and payments arrive on days of their own, so for a single day they are not on one basis. They are given in the month's report.

5 Tamarind Cross

Tamarind Cross's energy, plant, supply; commissioned 2023-03-01.

Energy

RENEWABLE FRACTION, TAMARIND CROSS
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

SITE LOSS, TAMARIND CROSS
8.1% • D
OK · Derived · gridintel.site.reference 0.1.0
Energy generated that reached no customer meter, as a share of energy generated.

DISTRIBUTION LOSS, TAMARIND CROSS
7.0% • D
OK · Derived · gridintel.site.reference 0.1.0
Share of energy sent out.

Energy flow, Tamarind Cross

From generation to customer meters, top to bottom. Each line keeps its own origin: a residual is shown as one.

	IN THE CHAIN	ENERGY
PV generation	Adds	404 kWh • C
Generator generation	Adds	0 kWh • C
Energy generated	Equals	404 kWh • C
Net into storage Energy into the battery less energy out of it: storage loss and any change in what is stored.	Less	-2 kWh • C
Station use A residual: the powerhouse's own use and anything unaccounted for.	Less	7 kWh • D
Energy sent out Metered where the powerhouse hands over to the network: the energy available to customers.	Equals	399 kWh • C
Distribution loss A residual: technical loss and unmetered use together.	Less	28 kWh • D
Energy delivered What customer meters recorded.	Equals	371 kWh • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

465 of 465 connections have a meter that recorded the whole period; 0 are unmetered and 0 incomplete.

Solar PV

PERFORMANCE RATIO, TAMARIND CROSS
71.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Counts curtailment as a loss, as IEC 61724 does.

PERFORMANCE RATIO BEFORE CURTAILMENT, TAMARIND CROSS
79.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Adds the curtailment estimate back: what the array could do.

PV CURTAILED, TAMARIND CROSS
10.1% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Of the PV energy that was available.

IN-PLANE IRRADIATION, TAMARIND CROSS
5.7 kWh/m² • C
OK · Calculated · gridintel.site.reference 0.1.0
Measured on site; the input to the performance ratio.

SPECIFIC YIELD, TAMARIND CROSS
4.0 kWh/kWp • C
OK · Calculated · gridintel.site.reference 0.1.0

CURTAILMENT, TAMARIND CROSS
46 kWh ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0
The controller's estimate. Not subtracted from metered generation, which is already net of it.

Battery (240 kWh, 80 kW, lithium iron phosphate)

EQUIVALENT FULL CYCLES, TAMARIND CROSS
0.6 • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy out of the battery over its nameplate capacity.

ROUND-TRIP EFFICIENCY, TAMARIND CROSS
89.1% ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0

STORAGE LOSS, TAMARIND CROSS
15 kWh ≈ D
Estimated inputs · Derived · gridintel.site.reference 0.1.0
The change in stored energy is estimated from state of charge and nameplate capacity.

STATE OF CHARGE, MEAN, TAMARIND CROSS
68.1% • C
OK · Calculated · gridintel.site.reference 0.1.0

STATE OF CHARGE, LOWEST, TAMARIND CROSS
40.9% • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY INTO STORAGE, TAMARIND CROSS
142 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY OUT OF STORAGE, TAMARIND CROSS
144 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

Generator (50 kVA, 40 kW, diesel)

FUEL PER KWH, TAMARIND CROSS
not computable ∅
Not computable · Calculated · gridintel.site.reference 0.1.0
Litres of fuel for each kWh the generator made.

RUNNING HOURS, TAMARIND CROSS
0.0 h • C
OK · Calculated · gridintel.site.reference 0.1.0
The advance of the hour meter over the period.

FUEL USED, TAMARIND CROSS
0 L • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR GENERATION, TAMARIND CROSS
0 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR'S SHARE OF ENERGY, TAMARIND CROSS
0.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

Supply

AVAILABILITY, TAMARIND CROSS
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

HOURS OF SUPPLY PER DAY, TAMARIND CROSS
24.0 h/day • C
OK · Calculated · gridintel.site.reference 0.1.0

INTERRUPTIONS PER CONNECTION, TAMARIND CROSS
0.0 • C
OK · Calculated · gridintel.site.reference 0.1.0

AVAILABILITY WITHOUT PLANNED INTERRUPTIONS, TAMARIND CROSS
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

Interruptions, Tamarind Cross

Every interruption recorded at the site in the period, in order.

> No interruption is recorded at Tamarind Cross in the period.

Sales and collection: not available for a day. Not available for a day. Collection efficiency and revenue per connection compare money billed and money collected over one span. Bills for a month's supply are not issued on the day the energy is used, and payments arrive on days of their own, so for a single day they are not on one basis. They are given in the month's report.

6 Method, sources and how to reproduce this report

Where every figure came from, and how a reader can check that this document is the one the records give.

Methodologies

METHODOLOGY	VERSION	WHAT IT SAYS OF ITSELF
GridIntel reference site performance `gridintel.site.reference`	`0.1.0`	GridIntel reference calculation — not a regulatory figure.

Sources

- Synthetic mini-grid vending and billing (synthetic)
- Synthetic mini-grid meter data (synthetic)
- Synthetic mini-grid plant monitoring (synthetic)
- Synthetic mini-grid site register (synthetic)

> Every source above is synthetic: designed demonstration data, not records of a real organization.

How to reproduce it

1. Build the report with definition version 1, for 14 September 2026, from the Demonstration Mini-grid Operator (synthetic) records.
2. The data fingerprint must be `b0167ca08ac57457af3bea882205afb118f40ea88543b10d1bbe5fea52b64234`. SHA-256 over the sorted SHA-256 digests of every distinct result the report's services were given by the repositories, each written as JSON with its keys in order.
3. With the same definition, methodologies and fingerprint the report ID is the same: GI-MSP-2026-09-14-2C52143B. A different ID means something above differs.
4. The figures with their status, origin, unit and method are in the CSV of the same report ID: gridintel-site-performance-2026-09-14-SAMPLE-SYNTHETIC.csv.

7 Regulatory references

> How far this report carries the fields of the instruments below, as read from the instruments themselves. It is not a claim that the report conforms to any regulator's or programme's template: none has been checked against the template it would be filed on.

INSTRUMENT	REQUIREMENT	THIS REPORT
NERC Mini-Grid Regulations 2023 (NERC-R-117-2023), s. 13(2) and s. 21(2)	Reports in the form prescribed in Schedule 4, per mini-grid or combined for a portfolio	**NOT CARRIED** Schedule 4 is not in the published text read. This is the operator's own performance report, not a filing.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Renewable energy fraction	**CARRIED** Sections 1 and each site's section.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Percentage of time of available power supply at a site per month, as measured by inverter output	**PARTLY** Each site's section gives customer-weighted availability from interruption records, not from inverter output.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Average consumption per user (ACPU); ARPU	**PARTLY** Each site's section gives consumption, energy sold and revenue per connection, for a month.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	OPEX, EBITDA	**NOT CARRIED** Not held.

8 Glossary and key

Energy sent out	Energy metered where the powerhouse hands over to the network: what is available to customers.
Station use	A residual: the powerhouse's own use and anything unaccounted for.

Site loss	Energy generated that reached no customer meter, as a share of energy generated.
Curtailment	PV energy that was available but not taken, as the controller estimates it.
Equivalent full cycles	Energy out of the battery over its nameplate capacity.
Specific yield	PV energy per kWp of array rating.
Fuel per kWh	Litres of fuel for each kWh the generator made.
Availability	Customer-weighted share of the period with supply: 1 − (customer-hours interrupted ÷ customer-hours in the period). Every recorded interruption is counted.
Hours of supply a day	The average hours a day the average connection had supply in the period.
Connection	A customer connection in the site's register. An active connection is one billed in the period.
Energy sold	Energy billed. For a prepaid account it is energy purchased, which may be used later.
Consumption	Energy recorded by customer meters.
Renewable fraction	The share of energy generated that came from PV.
Performance ratio	PV energy delivered against what the array's rating and the measured irradiation would give; curtailment counts as a loss (IEC 61724).
Collection efficiency	Revenue collected in the period ÷ revenue billed in the period, on a cash basis.
Claim	A figure a site's own report states. It agrees when it is on the basis of the calculation and within the tolerance of it.
Tariff assumption	A rate used to price bills in this dataset that is not a published or approved tariff.
Measured	Read from a meter, a device or a source record.
Reported	A figure published by someone else, shown as they stated it. It is never altered.
Calculated	Computed by GridIntel from records, under a named and versioned methodology.
Estimated	An estimate, not a measurement; the method is named with it.
Derived	A residual: what is left when other figures are subtracted. It carries their uncertainty.
Insufficient data	An input the figure needs is missing. No value is given rather than a guess, and never a zero.
Not available	No source holds this figure, or the data does not cover the period.
Same basis	A reported figure and a calculation count the same things over the same period by the same rule. Only figures on the same basis are compared.
Data fingerprint	A SHA-256 digest of every record the report was built from. The same records give the same fingerprint; any change to them gives another.
Report ID	The report's code, its period and the first characters of a digest of its definition, methodologies and data fingerprint. The same report of the same data has the same ID.

The marks beside a figure

• **OK**	Every input is present and measured.
≈ **Estimated inputs**	Calculated; every input is present, but at least one was estimated rather than measured.
! **Insufficient data**	An input this figure needs is missing. No value is shown rather than a guess.
∅ **Not computable**	The inputs exist but the calculation is undefined for them.
– **Not available**	The platform has no source for this figure.
M **Measured**	Read from a meter, device or source record.
R **Reported**	Published by someone else; shown as stated.
C **Calculated**	Computed by GridIntel under a named methodology.
E **Estimated**	An estimate, not a measurement.
D **Derived**	A residual: what is left after subtracting other figures. It inherits their uncertainty.

SAMPLE: SYNTHETIC DATA. End of report GI-MSP-2026-09-14-2C52143B.